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Oefeningenreeks 3D nr. 2.3

$f(x) = -2x^2 - 5x + 3$

$f'(x) = -4x - 5$

$f'(x) = 0$

Dan:

$-4x - 5 = 0$

$-4x = 5$

$x = -\dfrac{5}{4}$

$$f\left(-\frac{5}{4}\right) = -2 \cdot \left(-\frac{5}{4}\right)^2 - 5\left(-\frac{5}{4}\right) + 3 = -2 \cdot \frac{25}{16} + \frac{25}{4} + 3 =$$
$$= -\frac{50}{16} + \frac{100}{16} + \frac{48}{16} = \frac{98}{16} = \frac{49}{8}$$

| $x$ | $-\infty$ | $-\frac{5}{4}$ | $+\infty$ |
|---|---|---|---|
| $f'(x)$ | $+$ | $0$ | $-$ |
| $f(x)$ | $\nearrow$ | $\frac{49}{8}$ | $\searrow$ |

## References